

Cyberwarfare

Cyberwarfare is the use of cyber attacks against an enemy [state](#), causing comparable harm to actual [warfare](#) and/or disrupting vital [computer systems](#).^[1] Some intended outcomes could be [espionage](#), [sabotage](#), [propaganda](#), [manipulation](#) or [economic warfare](#).



Cyberwarfare specialists of the [United States Army's 782nd Military Intelligence Battalion \(Cyber\)](#) supporting the [3rd Brigade Combat Team, 1st Cavalry Division](#) during a [training exercise](#) in 2019

There is significant debate among experts regarding the definition of cyberwarfare, and even if such a thing exists.^[2] One view is that the term is a misnomer since no cyber attacks to date could be described as a war.^[3] An alternative view is that it is a suitable label for cyber attacks which cause physical damage to people and objects in the real world.^[4]

Many countries, including the [United States](#), [United Kingdom](#), [Russia](#), [China](#), [Israel](#), [Iran](#), and [North Korea](#),^{[5][6][7][8]} have active cyber capabilities for offensive and defensive operations. As states explore the use of cyber operations and combine capabilities, the likelihood of physical confrontation and violence playing out as a result of, or part of, a cyber operation is increased. However, meeting the scale and protracted nature of war is unlikely, thus ambiguity remains.^[9]

The first instance of [kinetic military action](#) used in response to a cyber-attack resulting in the loss of human life was observed on 5 May 2019, when the [Israel Defense Forces](#) targeted and destroyed a building associated with an ongoing cyber-attack.^{[10][11]}

Definition

There is ongoing debate over how cyberwarfare should be defined and no absolute definition is widely agreed upon.^{[9][12]} While the majority of scholars, militaries, and governments use

definitions that refer to state and state-sponsored actors,^{[9][13][14]} other definitions may include non-state actors, such as terrorist groups, companies, political or ideological extremist groups, [hacktivists](#), and transnational criminal organizations depending on the context of the work.^{[15][16]}

Examples of definitions proposed by experts in the field are as follows.

'Cyberwarfare' is used in a broad context to denote interstate use of technological force within computer networks in which information is stored, shared, or communicated online.^[9]

Raymond Charles Parks and David P. Duggan focused on analyzing cyberwarfare in terms of computer networks and pointed out that "Cyberwarfare is a combination of computer network attack and defense and special technical operations."^[17] According to this perspective, the notion of cyber warfare brings a new paradigm into military doctrine. [Paulo Shakarian](#) and colleagues put forward the following definition of "cyber war" in 2013, drawing on [Clausewitz](#)'s definition of war: "War is the continuation of politics by other means":^[13]

Cyber war is an extension of policy by actions taken in cyber space by state or nonstate actors that constitute a serious threat to a nation's security or are conducted in response to a perceived threat against a nation's security.

Taddeo offered the following definition in 2012:

The warfare grounded on certain uses of ICTs within an offensive or defensive military strategy endorsed by a state and aiming at the immediate disruption or control of the enemy's resources, and which is waged within the informational environment, with agents and targets ranging both on the physical and non-physical domains and whose level of violence may vary upon circumstances.^[18]

Robinson et al. proposed in 2015 that the intent of the attacker dictates whether an attack is warfare or not, defining cyber warfare as "the use of cyber attacks with a warfare-like intent."^[12]

In 2010, the former US National Coordinator for Security, Infrastructure Protection and Counter-terrorism, [Richard A. Clarke](#), defined cyberwarfare as "actions by a nation-state to penetrate another nation's computers or networks for the purposes of causing damage or

disruption".^[14] The target's own cyber-physical infrastructure may be used by the adversary in case of a cyber conflict, thus weaponizing it.^[19]

Controversy of term

There is debate on whether the term "cyber warfare" is accurate. In 2012, [Eugene Kaspersky](#), founder of [Kaspersky Lab](#), concluded that "cyberterrorism" is a more accurate term than "cyberwar." He states that "with today's attacks, you are clueless about who did it or when they will strike again. It's not cyber-war, but cyberterrorism."^[20] [Howard Schmidt](#), former Cyber Security Coordinator in the [Obama administration](#), said that "there is no cyberwar... I think that is a terrible metaphor and I think that is a terrible concept. There are no winners in that environment."^[21]

Some experts take issue with the possible consequences linked to the warfare goal. In 2011, Ron Deibert, of Canada's [Citizen Lab](#), warned of a "militarization of cyberspace", as militaristic responses may not be appropriate.^[22] However, to date, even serious cyber-attacks that have disrupted large parts of a nation's electrical grid (230,000 customers, [Ukraine, 2015](#)) or affected access to medical care, thus endangering life (UK [National Health Service, WannaCry, 2017](#)) have not led to military action.^[23]

In 2017, Oxford academic Lucas Kello proposed a new term, "Unpeace", to denote highly damaging cyber actions whose non-violent effects do not rise to the level of traditional war. Such actions are neither warlike nor peace-like. Although they are non-violent, and thus not acts of war, their damaging effects on the economy and society may be greater than those of some armed attacks.^{[24][25]} This term is closely related to the concept of the "grey zone", which came to prominence in 2017, describing hostile actions that fall below the traditional threshold of war.^[26] But as Kello explained, technological unpeace differs from the grey zone as the term is commonly used in that unpeace by definition is never overtly violent or fatal, whereas some grey-zone actions are violent, even if they are not acts of war.^[27]

Cyberwarfare vs. cyber war

The term "cyberwarfare" is distinct from the term "cyber war". Cyberwarfare includes techniques, tactics and procedures that may be involved in a cyber war, but the term does not imply scale, protraction or violence, which are typically associated with the term "war", which inherently refers to a large-scale action, typically over a protracted period of time, and may include objectives seeking to utilize violence or the aim to kill.^[9] A cyber war could accurately